Vikram Sandu

Contact Information

Address	Home No 411, Charano ka Bass Borunda, Jodhpur, Rajasthan, India - 342604
Email	sanduvikram95@gmail.com
Phone	+91 9116995878
Webpage	https://vikramsandu.github.io/

Research Interests

- 3D Computer Vision
- Image and Video Signal Processing
- Machine Learning

Education

M.Tech - Indian Institute of Technology (IIT), Hyderabad Aug 2017 - May 2019
Dept of Electrical Engineering
- Thesis : 3D Reconstruction of the Human Face using Multi-stereo Camera Network
- Advisor : Dr. Soumya Jana
- CGPA : 9.06
- Recipient of MHRD scholarship (Govt of India)

B.Tech - University College of Engineering and Technology, Bikaner Aug 2012 - May 2016
Dept of Electronics and Communication Engineering
- Percentage: : 65
- GATE Score : 658 (AIR 1550)

Course Work

Mathematics	: Linear Algebra, Probability and Random Process, Linear and Non-Linear Optimization
Electrical	: Practical Challenges in Image Analysis, Optimization Methods in Machine Learning, Bayesian Data Analysis, Introduction to AI and ML

Computer Skills

Programming Languages	: Python, C, MATLAB
Platforms	: Windows, LINUX
Softwares	: $\LaTeX$

Industrial Experience

Checko, IIT Kanpur Sep 2021 - Jan 2024
- Role : Computer Vision Engineer
- Advisor : Prof. Deepak Gupta, IIT Kanpur
- Worked on developing anti-counterfeiting technologies, optimizing the app to be smaller and faster while maintaining its ability to detect counterfeit items.

Updated on April 9, 2025

Tata Consultancy Services, Hyderabad Aug 2019 - Sep 2021

- ◇ Role : Software Engineer
- ◇ Worked on diverse projects across areas such as Machine Learning, AngularJS, Amazon Web Services (AWS), Power BI, and more.

LVPEI, Hyderabad Aug 2018 - Oct 2018

- ◇ Role : Intern
- ◇ Team : Research and Development
- ◇ Developed a patient-friendly multi-camera system for facial image acquisition and 3D reconstruction, enabling pre-operative planning and post-operative assessment for facial surgeries.

Academic Experience

Indian Institute of Science (IISc) June 2024 - Present

- ◇ Role : Research Associate
- ◇ Lab : Vision and Image Processing (VIP)
- ◇ Advisor : Dr. Rajiv Soundararajan
- ◇ Working on dynamic novel view synthesis methods based on NeRF, K-Planes, and Gaussian Splatting, with a focus on improving Gaussian Splatting based methods for complex and long-range motion scenarios.
- ◇ A paper based on this work has been recently submitted to ICCV 2025 and is currently under peer review.

Indian Institute of Technology (IIT) Hyderabad Aug 2017 - May 2019

- ◇ Teaching Assistant
- ◇ Advanced Digital Signal Processing - Aug 2018 semester

Publications

- ◇ Thesis:
  - ➢ Vikram Sandu and Soumya Jana. 3d reconstruction of the human face using multi-stereo camera network. *IITH Archive*, 2019. [Link]
- ◇ Conferences:
  - ➢ Roopak Tamboli, Vikram Sandu, and Soumya Jana. Novel hybrid teleophthalmology: Technological case for oculofacial surgery. In *9th Annual IEEE Global Humanitarian Technology Conference, GHTC, 17-20 October 2019, Seattle, United States*, 2019. [Link]

Projects

Distributed optimization in Multi-agent System.

- ◇ Course Project for "Optimization Methods in ML"
- ◇ This project was aimed to optimize a global function formed by a sum of local functions, using only local communication and local computation. Building upon Nesterov gradient descent, We proposed a weight sharing strategy among the local agents that leads to a better convergence rate and sub-optimality.

CommonLit - Evaluate Student Summaries

- ◇ Kaggle Competition
- ◇ In this Kaggle competition, the objective was to train a language model to assess summaries written by students in grades 3 to 12 for a given prompt.
- ◇ I proposed a simple Data pre-processing approach to incorporate contextual information from the prompt text into summaries before feeding them into the transformers. We got bronze medal in this competition. [Link]

Research paper Implementations and Benchmarking

- Implemented and benchmarked several research papers from scratch in PyTorch.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. [Link]
- Prafulla Dhariwal and Alex Nichol. Diffusion models beat gans on image synthesis. *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. [Link]
- Chien-Yao Wang, Alexey Bochkovskiy, and Hong-Yuan Mark Liao. Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *arXiv preprint*, 2022. [Link]
- Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. [Link]
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Link]

Language Modeling: A Digestible Walkthrough

- Medium Blog
- In this blog, I have consolidated insights from various sources on language modeling, presenting them in a clear and structured manner for easier understanding. [Link]